

1. Data Splitting Strategy

Split Ratio: 80% Training, 20% Testing.

Why Split? We hide 20% of the data from the model during training. This allows us to test the model on "unseen" data to verify if it can generalize to new patients, rather than just memorizing the old ones.

2. Model Performance Metrics

Accuracy: ~75-80% (varies slightly by run). This means the model correctly predicted the outcome for about 80% of patients.

Precision: Measures how trustworthy a "Positive" prediction is. If the model says a patient has heart disease, how often is it right?

Recall (Sensitivity): Measures the model's ability to find all actual cases. Out of all patients who actually have heart disease, how many did we catch?

3. Confusion Matrix Interpretation The matrix shows four types of results:

True Positives (TP): Correctly predicted heart disease.

True Negatives (TN): Correctly predicted healthy.

False Positives (FP): Predicted heart disease, but patient is healthy (False Alarm).

False Negatives (FN): Predicted healthy, but patient has heart disease (Missed Case - Dangerous!).

4. Conclusion The Logistic Regression model performs reasonably well. For medical diagnosis, we typically aim for higher Recall to minimize "False Negatives," ensuring we don't miss any sick patients.